

Environmental Technology

ISSN: 0959-3330 (Print) 1479-487X (Online) Journal homepage: www.tandfonline.com/journals/tent20

Effect of circulation and micro-aeration on sludge characteristics and microbial community in an ABR for treating traditional Chinese medicine wastewater

Chengyuan Su, Qiujin Deng, Yuxiang Lu, Jiaqi Pan, Wuyang Chen, Shenglong Chen, Xue Deng, Xiangfeng Lin & Zhi Huang

To cite this article: Chengyuan Su, Qiujin Deng, Yuxiang Lu, Jiaqi Pan, Wuyang Chen, Shenglong Chen, Xue Deng, Xiangfeng Lin & Zhi Huang (2020) Effect of circulation and micro-aeration on sludge characteristics and microbial community in an ABR for treating traditional Chinese medicine wastewater, Environmental Technology, 41:25, 3284-3296, DOI: 10.1080/09593330.2019.1604818

To link to this article: https://doi.org/10.1080/09593330.2019.1604818

View supplementary material

Published online: 18 Apr 2019.

Submit your article to this journal

Article views: 408

View related articles

View Crossmark data

Citing articles: 5 View citing articles

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=tent20

ENVIRONMENTAL TECHNOLOGY
2020, VOL. 41, NO. 25, 3284–3296
https://doi.org/10.1080/09593330.2019.1604818

Taylor & Francis
Taylor & Francis Group

Check for updates

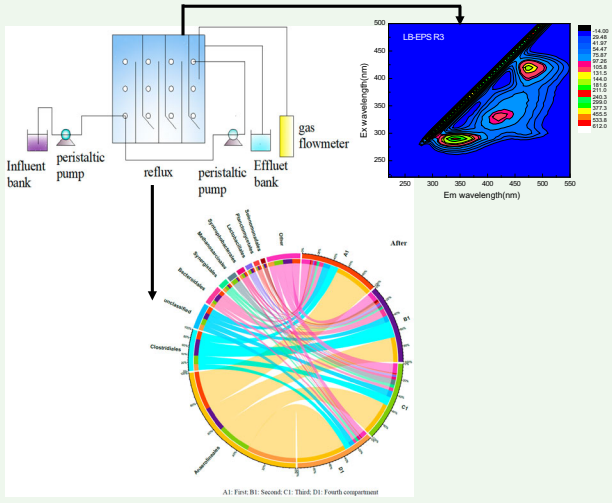
Effect of circulation and micro-aeration on sludge characteristics and microbial community in an ABR for treating traditional Chinese medicine wastewater

Chengyuan Su[a,b], Qiujin Deng[a], Yuxiang Lu[a], Jiaqi Pan[a], Wuyang Chen[a], Shenglong Chen[a], Xue Deng[a], Xiangfeng Lin[a] and Zhi Huang[a]

[a]Key Laboratory of Ecology of Rare and Endangered Species and Environmental Protection, Guangxi Normal University, Ministry of Education, Guilin, People's Republic of China; [b]University Key Laboratory of Karst Ecology and Environmental Change of Guangxi Province, Guangxi Normal University, Guilin, People's Republic of China

ABSTRACT

The effects of circulation reflux and micro-aeration on the performance of a modified anaerobic baffled reactor (ABR) for treatment of traditional Chinese medicine (TCM) wastewater were evaluated. The characteristics of anaerobic sludge and microbial community structure in the modified ABR were also investigated. The results indicated that with conditions of reflux ratio of 1, reflux ratio of 2, reflux ratio of 2 with micro-aeration, and reflux ratio of 3, the modified ABR achieved an average COD removal efficiency of 90%, 87.7%, 87.8%, and 88.4%, respectively. In addition, the NH_3-N average removal efficiency was 45.1%, 50%, 55.9%, and 55.4%, respectively. The analysis of excitation–emission matrix (EEM) fluorescence spectra of soluble microbial products (SMP) and extracellular polymeric substances (EPS) showed that there were tyrosine-like, aromatic protein-like, and coenzyme F_{420} substances in the sludge. The EPS were analysed by the Fourier transform infrared spectroscopy (FTIR), which showed that aromatic compounds were partially degraded, while the protein and polysaccharide compounds increased in each compartment of the modified ABR. Interestingly, the microbial community of anaerobic sludge analysis results showed that *Chloroflexi* was the dominant in the first, third and fourth compartments. Meanwhile, *Levilinea* and *Methanothrix* were the dominant species in the first and third compartments at the genus level.



ARTICLE HISTORY
Received 17 February 2019
Accepted 31 March 2019

KEYWORDS
Modified ABR; TCM wastewater; operational efficiency; soluble microbial products; microbial community compositions

1. Introduction

Traditional Chinese medicine (TCM) is an integral part of Chinese culture. In the medical field, TCM has displayed unique characteristics of efficacy and curative effects. In recent years, the TCM industry has been vigorously developed after encouragement and advocacy by the state. However, a large amount of wastewater is produced during the production of TCM through the

CONTACT Chengyuan Su suchengyuan2008@126.com Key Laboratory of Ecology of Rare and Endangered Species and Environmental Protection, Guangxi Normal University, Ministry of Education, 15 Yucai Road, Guilin 541004, People's Republic of China; University Key Laboratory of Karst Ecology and Environmental Change of Guangxi Province, Guangxi Normal University, 15 Yucai Road, Guilin 541004, People's Republic of China

Supplemental data for this article can be accessed at https://doi.org/10.1080/09593330.2019.1604818.

© 2019 Informa UK Limited, trading as Taylor & Francis Group

process of washing, soaking, and decoction of medicinal materials, among which the main components are polysaccharides, cellulose, protein and some toxic substances. People also need to address the pollution problems caused by TCM wastewater [1]. The characterization of TCM wastewater is high concentrations of organic matter, high suspended substance, complex composition, large variation of wastewater quality, etc. Therefore, some scholars have begun to explore the potential treatment processes of TCM wastewater. Considering the characteristics of TCM wastewater, advanced oxidation processes (AOPs), membrane treatment method, advanced oxidation-aerobic biological process, and anaerobic-anoxic-aerobic combined process were employed [2–4].

In comparison to physicochemical technologies, anaerobic processes have several advantages including the lack of need for aeration (and thus less energy requirement), low energy consumption, biogas production, and withstanding of high organic-loading [5–6]. High efficiency anaerobic baffled reactor (ABR) processes have an important influence on the development of modern anaerobic biological treatment technology. ABR has been extensively used in the treatment of pulp and paper mill black liquors, tannery, textile dye, brewery, and domestic wastewaters [7–10]. Researchers' continue to work to improve and optimize ABR forms of different types to achieve different treatment objectives, with the ultimate goal of improving the ability of ABR's retention of high biomass concentrations and even distribution of influent. These modified methods include the use of carriers to support the growth of microorganisms, using a larger number of compartments, and the addition reflux [11]. Their results showed that reflux could degrade toxic substances in wastewater, alleviate inhibitory effects on the ABR, and resist high organic load of the influent. At the same time, the circulation reactor significantly improved sludge distribution and mass transfer, and allowed for the direct contact of microorganisms with the influent in each compartment [12]. Some scholars have also studied the effects of adding micro-aeration to the ABR and expanded granular sludge bed (EGSB) reactor [13]. Liu and Dong reported that the micro-aerobic granular sludge reactors had stable COD removal effect under different operational modes, whereas nitrogen removal depended on oxygen influx [13]. To our knowledge, few scholars have combined these two improvement methods to improve the treatment efficiency; especially the effects of reflux and micro-aeration on the sludge characteristics and microbial community in ABR have few investigated.

Our modified ABR was designed to increase reflux and micro-aeration. The performance of a modified ABR for treatment of TCM wastewater was investigated. In addition, the soluble microbial products (SMP) and extracellular polymeric substances (EPS) of the anaerobic sludge was measured by excitation–emission matrix (EEM) and Fourier transform infrared spectroscopy (FTIR). Moreover, the microbial community structure in modified ABR was explored by 16S rRNA high-throughput sequencing.

2. Materials and methods

2.1. *ABR configuration*

The schematic diagram of the modified ABR is shown in Fig. S1. The ABR was made of polymethyl methacrylate. The total volume was 12 L, and the working volume was 10 L with length×width×height of 400 mm×100 mm×300 mm. It was divided into four compartments included an up-flow zone and a down-flow zone with a volume ratio of 3:1. The reactor was sealed and a gas gathering system was placed in the top of each compartment. The up-flow zone was equipped with three sampling ports, which was convenient for collecting samples. The influent was fed to the reactor compartments by a peristaltic pump (Longer WT600-2, Longer pump Co., China) [10]. There was no reflux and micro-aeration device at the start-up stage. After the start-up stage, in order to improve the performance of the reactor for treatment of real TCM wastewater, a peristaltic pump was added at the sampling port above the fourth compartment to connect with the first compartment to form a reflux device. In addition, a micro-aeration device was added in the fourth reactor compartment. The aeration process was controlled by a gas flowmeter at dissolved oxygen concentration of 0.5–1.0 mg/L.

2.2. *Experimental water and inoculation sludge*

In this experiment, wastewater was collected from a TCM plant in Guilin, China. The primary characteristics of this TCM wastewater were shown in Table 1. During the experiments, the COD concentration of the influent was controlled to approximately 2000mg/L. $NaHCO_3$ and nutrient solution were added in the influent, and the pH value of the influent was kept

Table 1. The characteristics of the TCM wastewater.

Parameter	COD (mg/L)	SCOD (mg/L)	SS (mg/L)	NH_3-N (mg/L)	pH
Concentrations	1900–2900	1300–2100	150–220	70–140	4.0–5.5

within values ranging 6.5–7.5. The inoculated sludge was obtained from a brewery in Guilin city, and the total suspended solids (TSS) of anaerobic sludge were 20.8 g/L. The ratio of volatile suspended solid (VSS) to TSS was 0.74.

2.3. Reactors operation

This study was roughly divided into the following three stages: sludge cultivation, sludge domestication, and improved treatment of TCM wastewater stages. The stage of sludge domestication culture was divided into three phases: the first phase was only synthetic glucose wastewater (A), the second phase was synthetic glucose wastewater and real TCM wastewater as the composition ratio was 1:1 (B), and the third phase was total real TCM wastewater, which each phase ran for 15–20 days and COD concentration was controlled around 2000–2500 mg/L. The stage of improved treatment was divided into four phases: reflux ratio of 1 (C), reflux ratio of 2 (D), reflux ratio of 2 with micro-aeration (E), and reflux ratio of 3 (F). The ABR was operated at hydraulic residence time (HRT) 24 h and room temperature 20–25°C.

2.4. Analytical methods

During the experiment, the removal of COD and NH_3-N was analysed on each day. SMP and EPS of the sludge in the reactor were also investigated in the different stages. The sludge was extracted at 45 d (stage C) and 80 d (stage E). The sludge was harvested following centrifugation at 8000 rpm for 10 min; the resulting supernatant represented the SMP [14]. The sludge pellet in the tube was then resuspended into 15 mL of 0.05% NaCl, sonicated at 20 KHz for 2 min, shaken horizontally at 150 rpm for 10 min, and sonicated again for an additional 2 min [15]. The liquor was then centrifuged at 8000 rpm for 10 min to separate solids from the supernatant. The collected supernatant was regarded as the loosely bound EPS (LB-EPS). The residual sludge pellet left in the centrifuge tube was resuspended in a 0.05% NaCl solution, sonicated for 3 min, then heated at 80°C for 30 min in a water bath, and finally centrifuged at 8000 rpm for 10 min to collect the tightly bound EPS (TB-EPS) [16]. The names of the samples were 'SMP 1-4,' 'SMP R1-R4,' 'LB-EPS 1-4,' 'LB-EPS R1-R4,' 'TB-EPS 1-4,' and 'TB-EPS R1-R4,' respectively. The analysis method of EPS was heating extraction. Based on the identification of fluorescent compounds, the EEM technique was adopted to further analyse SMP and EPS components using a fluorescence spectrophotometer (F-7000, Hitachi, Japan) [17]. In this study, the EEM fluorescence spectra were collected with subsequent scanning at emission wavelengths from 220 to 600 nm at 5 nm slits, and excitation wavelengths from 220 to 500 nm also at 5 nm slits. In addition, the major functional groups in sludge SMP and EPS were analysed by a FTIR spectrometer (Thermo Nicolet, USA).

High-throughput sequencing was used to analyse the microbial community variation under different conditions. The sludge samples from each compartment were collected and mixed to a composite sample for bacterial DNA extraction. The qualified DNA was sequenced through high-throughput pyrosequencing on a Miseq sequencing platform. The polymerase chain reaction (PCR) amplification of the hypervariable regions used V3-V4 universal primers 341F (CCTACGGGNGGCWGCAG) and 805R (GACTACHVGGGTATCTAATCC) for bacteria [18,19]. The sequence was compared with the reference microorganisms available in Ribosomal Database Project (RDP) (http://rdp.cme.msu.edu/misc/resources.jsp), Silva (http://www.arb-silva.de), and NCBI 16S database (http://ncbi.nlm.nih.gov). In order to obtain the species classification information corresponding to *each operational taxonomic units* (OUT), the RDP Bayesian classifier algorithm was used for taxonomic analysis of 97% similarity with OTU representative sequences [18,19]. Network analysis was performed using QIIME, and Shannon diversity indices were calculated for each sample. In the NCBI Sequence Read Archive database, the biosample accession number of bacteria was SAMN10410857.

3. Results and discussion

3.1. Removal efficiency of COD and NH_3-N

TCM wastewater usually contains high organic contaminants. The operation stability and performance of anaerobic reactors were usually evaluated by detecting the removal efficiency of COD. The detection result of removal COD is shown in Figure 1. At stage A, the influent COD concentration was controlled at 2000–2500 mg/L with a glucose substrate. At 1–15 days, the removal efficiency of COD increased rapidly, and it reached 93% on average at 15–20 days. This indicated that anaerobic sludge has an adaptive process during the inoculation stage, and the removal efficiency of COD tends to increase over time, and could be maintained at more than 90%. As shown in Figure 1, after the wastewater treatment entered into the acclimatization stage, the COD concentration of the influent was controlled at 1500–2300 mg/L as the glucose, real TCM wastewater substrate, and the COD removal efficiency was maintained at 93–95% in the stage B. During the

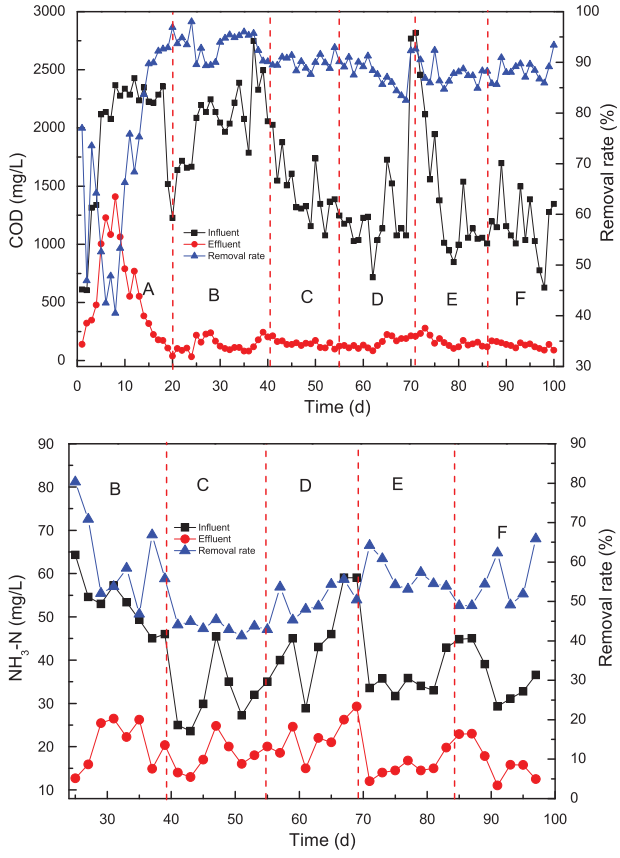


Figure 1. Removal efficiency of COD and NH_3-N at different conditions.

stages C, D, E, and F, the real TCM wastewater composition became more complicated, and the COD concentration in the influent was volatile when controlled between 1000–2000mg/L. At stage C, the reflux device was started. When the reflux ratio was 1, the removal efficiency of COD was stable between 88% and 93%, and the average removal efficiency of COD was 90%. This result indicated that the sludge in the anaerobic reactor had been successfully acclimatized from stage B to stage C, and the treatment effect was still functioning correctly. When the reflux ratio was increased to 2 at stage D, the average removal efficiency of COD was about 89%. However, at the 64th to 69th days, the removal efficiency of COD reduced to 82.5%, maybe there were complex and some bio-recalcitrant organic compositions, phenols, humic acid, etc [1]. in the TCM wastewater. When the reactor ran to a certain stage, toxic substances might accumulate in the anaerobic reactors. In addition, with reflux ratio increasing, some loose sludge was washed away from the reactor, resulting in the SS concentration increasing in the effluent [20]. SS in the effluent contributes a part of COD, leading to the increase of COD concentration in the effluent. When the sludge adapted to the reactor, the removal efficiency of COD increased back to more than 92%. At stage E, the COD removal efficiency remained between 85% and 90% with the reflux ratio 2 and additional aeration. Compared with stage D, the removal efficiency of COD had a small decline. The metabolic activity of methanogen in the last compartment of the ABR was influenced by the addition of a micro aeration device. Methanogens are very sensitive and need to adapt to changing environment. At stage F, the removal efficiency of COD was still maintained between 85% and 90% with reflux ratio 3. Through the whole process, the removal efficiency of COD was always maintained within the range of 85% to 93%, indicating that the removal efficiency of COD was no significant effect by the modified ABR with additional reflux ratio and micro aeration.

In this experiment, the NH_3-N removal efficiency was also discussed (Figure 1). Because of the composition of TCM wastewater, the NH_3-N concentration could not be controlled stably. At stage B, glucose and real TCM wastewater constituted the majority of the influent. The NH_3-N removal efficiency was about 53%. The removal efficiency of NH_3-N in stage C was between 40% and 50%, and the average removal efficiency of NH_3-N was 45%, which indicated that the removal efficiency of NH_3-N did not increase obvious at the reflux ratio of 1. Compared with only real TCM wastewater, the composition of glucose provides a biodegradable carbon source which is why the removal efficiency of NH_3-N was higher. In stage D with a reflux ratio of 2, the average removal efficiency of NH_3-N reached to 50%. After increasing micro-aeration in stage E, the average removal efficiency of NH_3-N increased to 56%, and the removal efficiency was between 53% and 65%. The NH_3-N concentration was kept between 12–20 mg/L in the effluent during this time. Increasing the amount of dissolved oxygen in the influent after micro-aeration allows the ammonia nitrogen to convert to nitrite nitrogen by nitrifying bacteria under aerobic environment of the modified ABR. Nitrification process was present in the reactor [21]. In stage F, the NH_3-N removal efficiency ranged from 49% to 66%. Here, it was illustrated that the increasing reflux ratio and micro-aeration was benefited the removal efficiency of NH_3-N.

3.2. *Sludge characteristics*

Before and after the improvement, changes in the sludge characteristics were analysed in the modified ABR. Previous studies have reported considerable variability in the production of SMP in response to environmental stresses imposed by the microorganisms [14,15]. From

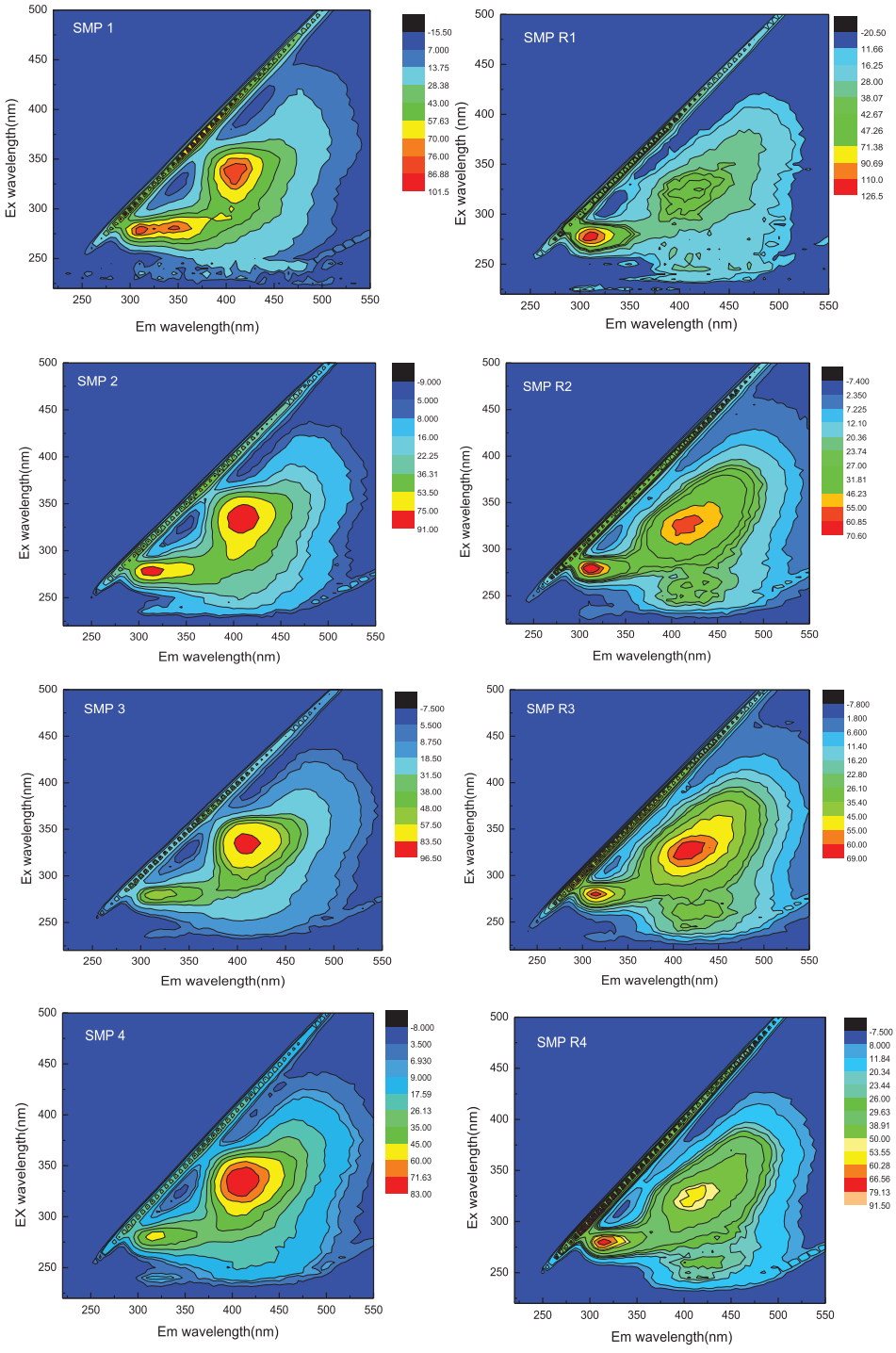


Figure 2. EEM fluorescence spectra of SMP in each compartment.

another perspective, the ingredients and proportions of proteins, polysaccharides, and other polymers in the EPS of anaerobic sludge were closely related to the microbial communities and their living environment. Additionally, the sludge sedimentation performance, physicochemical character, surface features of the sludge, and biological flocculation properties, were influenced by EPS [22–25]. Therefore, the SMP and EPS in each compartment of the reactor were analysed by EEM and FTIR spectra.

3.2.1. EEM fluorescence spectra analysis

EEM spectroscopy is an important technique for monitoring and analysing bioreactors. It has the characteristics of high sensitivity, requiring fewer samples, high accuracy, and good speed. As shown in Figure 2, before the

modification, the SMP of sludge in the reactor had three fluorescence peaks in the EEM spectra, which were of a type of tyrosine-like fluorescence peak (Ex/Em = 280/310 nm), fulvic acid-like fluorescence peak (Ex/Em = 325/350 nm), and humic acid-like fluorescence peak (Ex/Em = 340/410 nm). After improvement of reflux and micro-aeration, the SMP of the sludge mainly had four fluorescence peaks; tyrosine-like fluorescence peak (Ex/Em = 280/310 nm), humic acid-like fluorescence peak (Ex/Em = 340/410 nm), fulvic acid-like fluorescence peak (Ex/Em = 325/350 nm), and another class fulvic acid-like fluorescence peak (Ex/Em = 260/420 nm) in the EEM spectra of each compartment [20,26]. For the first compartment, the absorption peak intensity of tyrosine-like increased slightly, while the absorption peak intensity of humic acid-like decreased. The contents of humic acid-like were reduced in the first compartment to avoid acidification, and the contents of tyrosine increased in the first compartment of anaerobic sludge to promote the physiological activities of microorganisms in the reactor. The intensity of tyrosine-like and humic acid-like fluorescence absorption peaks decreased at a small range in the EEM spectra of the sludge in the second reactor compartment, which showed that after the modification, the contents of SMP decreased in the second compartment of the reactor. It was illustrated that the microbes consumed more protein to supply their own metabolism and the microbial activity was more active in the second compartment. The content of humic acid-like reduction could avoid acidification in the reactor with additional reflux ratio and micro aeration which made the reactor operate stably. The intensity of humic acid-like absorption peak in the EEM spectra of SMP reduced in the third compartment of the modified ABR. This result showed that the improvement reflux ratio could also reduce toxic substances in the third reactor compartment of the reactor. In EEM spectra of SMP, the absorption peak of tyrosine-like significantly increased, while the absorption peak of humic acid-like decreased in the fourth compartment of the modified ABR. The results showed that TCM wastewater was effectively treated in the fourth reactor compartment after the improvement reflux ratio. In conclusion, it was demonstrated that the visible fulvic acid-like absorption peak appeared, and visible humic acid-like absorption peak decreased, avoiding the acidification of the reactor in each compartment after the modification. This was consistent with the high removal rate of COD and NH_3-N in the modified ABR.

This was all except for the second compartment, where the content of tyrosine increased in each compartment of the reactor. Previous studies have shown that the formation of anaerobic granular sludge has a close relationship with tyrosine. Therefore, the contaminants in the TCM wastewater were effectively degraded and the modified ABR was stably operated. In Figure 3, before modification of reflux and micro-aeration, the LB-EPS mainly had five fluorescence peaks in the EEM spectra of each compartment, which was characterized by tyrosine-like (Ex/Em = 290/345 nm), fulvic acid-like (Ex/Em = 325/350 nm), coenzyme NADH (Ex/Em = 335/420 nm), another class fulvic acid-like (Ex/Em = 390/430 nm), and coenzyme F_{420} fluorescence peaks (Ex/Em = 420/475 nm) [20,21]. After modification, the intensity of the coenzyme F_{420} fluorescence absorption peak was significantly reduced, the intensity of the coenzyme NADH fluorescence absorption peak increased, and the intensity of the protein absorption peak increased slightly in amplitude during the first compartment of the reactor. It was shown in the treatment that methanogens decayed in the first compartment because of micro-aeration. The fluorescence intensity of fulvic acid-like and coenzyme F_{420} increased slightly, indicating that methane-producing activity was present in the second compartment. At the same time, the intensity of coenzyme NADH fluorescence absorption peak was significantly increased. The fluorescence intensity of protein-like also tended to increase. The increasing fluorescence intensity of coenzyme F_{420} indicated that the second compartment was in good operating condition. In the third compartment, the fluorescence peak of humic acid-like (Ex/Em = 300/470 nm) appeared. However, the humic acid disappeared after improvement. At the same time, the fluorescence intensity of protein and coenzyme NADH increased. In addition, the fluorescence intensity of coenzyme F_{420}, coenzyme NADH, fluvic acid-like, and protein-like increased significantly in the fourth compartment, especially those of coenzyme F_{420} and protein-like. The reason may be that in the fourth compartment of the reactor, addition of the aeration device created an environment where the strictly anaerobic microorganisms could not adapt to the environment and then declined in number, while the facultative anaerobe would enter a hypermetabolic state. On a whole, the microbial metabolism was vigorous in the fourth compartment after the modification. The toxic substances in TCM wastewater were degraded, which was conducive to the stable operation of the reactor. In other words, it was demonstrated that after the modification of reflux and micro-aeration, there was an increasing trend in the amount of protein-like substances and coenzyme NADH in each compartment. Therefore, the stability of sludge could be improved, which provided a guarantee for the operation of the modified ABR.

As shown in Fig. S2, the three main fluorescence peaks were amino acid-like (Ex/Em = 235/320 nm),

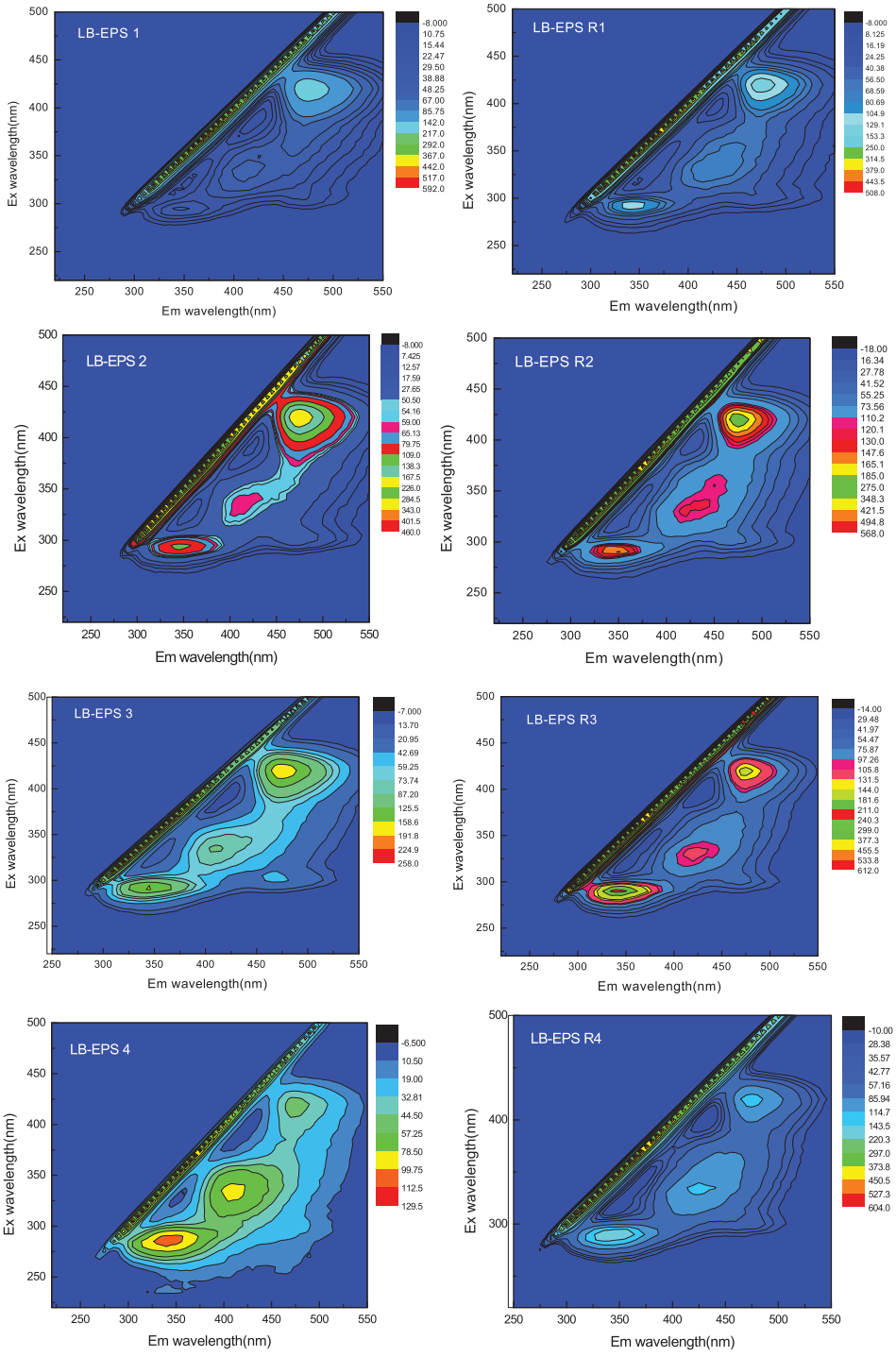


Figure 3. EEM fluorescence spectra of LB-EPS in each compartment.

tryptophan-like (Ex/Em = 290/350 nm), and fulvic acid-like (Ex/Em = 400/440 nm) peaks which appeared in the EEM spectra of TB-EPS without reflux and micro-aeration. After the modification, aromatic protein-like (Ex/Em = 230/320 nm), tryptophan (Ex/Em = 290/350 nm), fulvic acid-like (Ex/Em = 325/365 nm), and another class fulvic acid-like (Ex/Em = 325/420 nm) peaks were observed in the EEM spectra. Addition of reflux and micro-aeration decreased the fulvic acid-like fluorescence peak (Ex/Em = 400/440 nm) in the EEM spectra of the TB-EPS. Moreover, the fluorescence peak intensity of amino acid-like and tryptophan-like substances in each compartment significantly increased. The proportion of EPS components was important in influencing the characteristics of EPS [20]. Here it was illustrated that the sludge properties had changed to some extent in each compartment of the reactor.

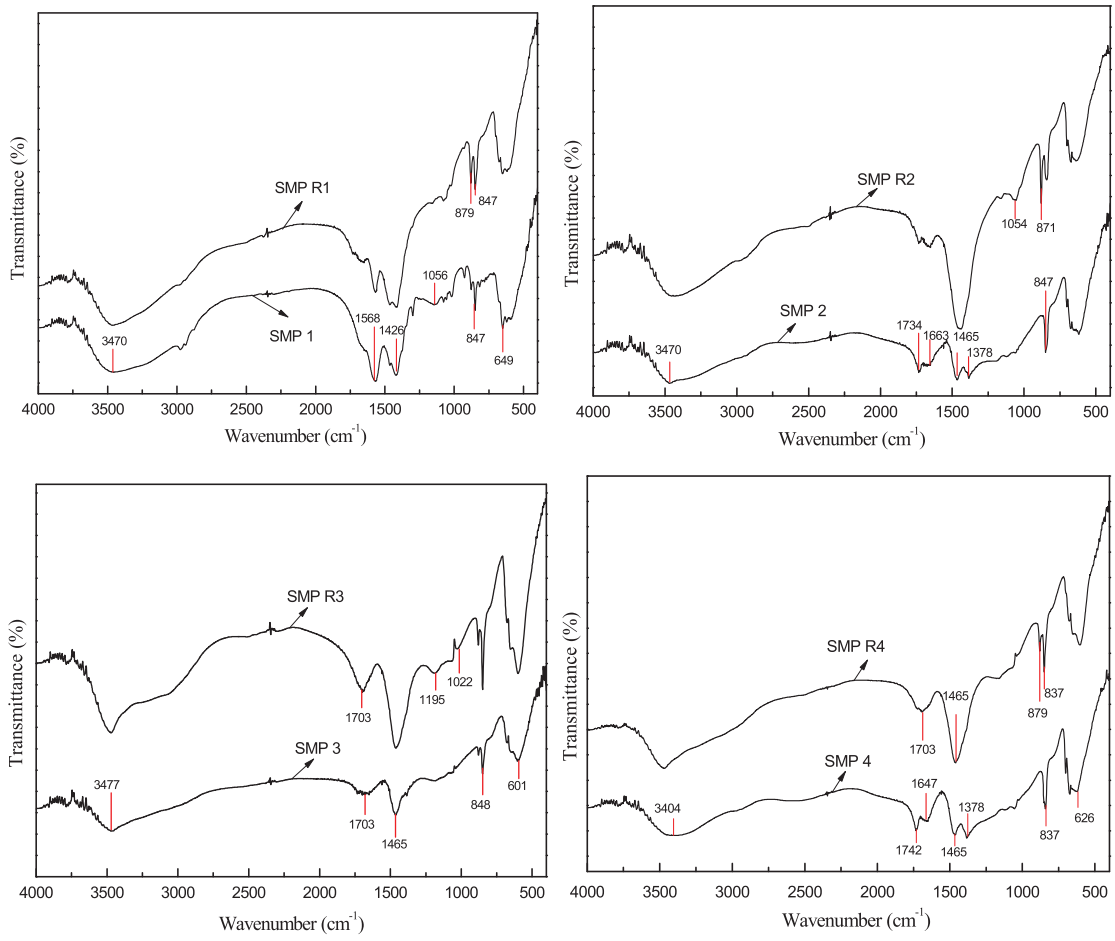


Figure 4. FTIR spectra of SMP in each compartment.

3.2.2. FTIR spectra analysis

FTIR spectroscopy has the advantages of high scanning speed, good reproducibility, and high signal-to-noise ratio. It is usually used to analyse the functional groups and molecular structures in SMP and EPS in order to determine the content of aromatic compounds, proteins, polysaccharides, and other substances contained in sludge [20,26]. In Figure 4, the absorption peak of 1426 cm^{-1} (C–N stretching vibration) was not significantly altered, indicating that aromatic compounds existed before and after the modification of reflux and micro-aeration in the SMP of the first compartment of the reactor. In the area of 1120–1020 cm^{-1}, the C–O–C stretching vibration of polysaccharides indicated that the polysaccharide content decreased in the first compartment by the modification [17]. For the second compartment, the absorption peak of 1456 cm^{-1} was obvious after the modification reflux and micro-aeration, indicating that the aromatic compounds increased significantly with the modification. And the absorption peak of bending vibration of the C–H bond appeared at 1120–949 cm^{-1} with a small amplitude enhancement [17,20]. It was indicated that the properties of granular sludge may be affected by the modification in the second reactor compartment. The intensity of the absorption peak of N-H stretching vibration at 3500–3300 cm^{-1} and the C=O stretching vibration at 1703 cm^{-1} were increased in the third compartment [27,28]. Meanwhile, the absorption peak of 1456 cm^{-1} significantly increased, indicating that aromatic compounds significantly increased. The absorption peak of C–H bending vibration appeared at 949–1120 cm^{-1}. It was indicated that the sludge morphology in the third compartment was affected to some extent by the modification. The N–H stretching vibration peak (3500–3300 cm^{-1}) changed from a broad peak to a sharp peak in the SMP of the sludge in the fourth compartment, which may be due to the complexation reaction between the protein in the wastewater and the metal ions of the sludge in the fourth compartment. The characteristic peaks of C=O stretching vibration at 1647 and 1742 cm^{-1} disappeared, and the characteristic peaks of C=O stretching vibration at 1703 cm^{-1} appeared [17]. Similarly to the third compartment, the characteristic absorption peak of 1456 cm^{-1} was significantly enhanced in the fourth compartment.

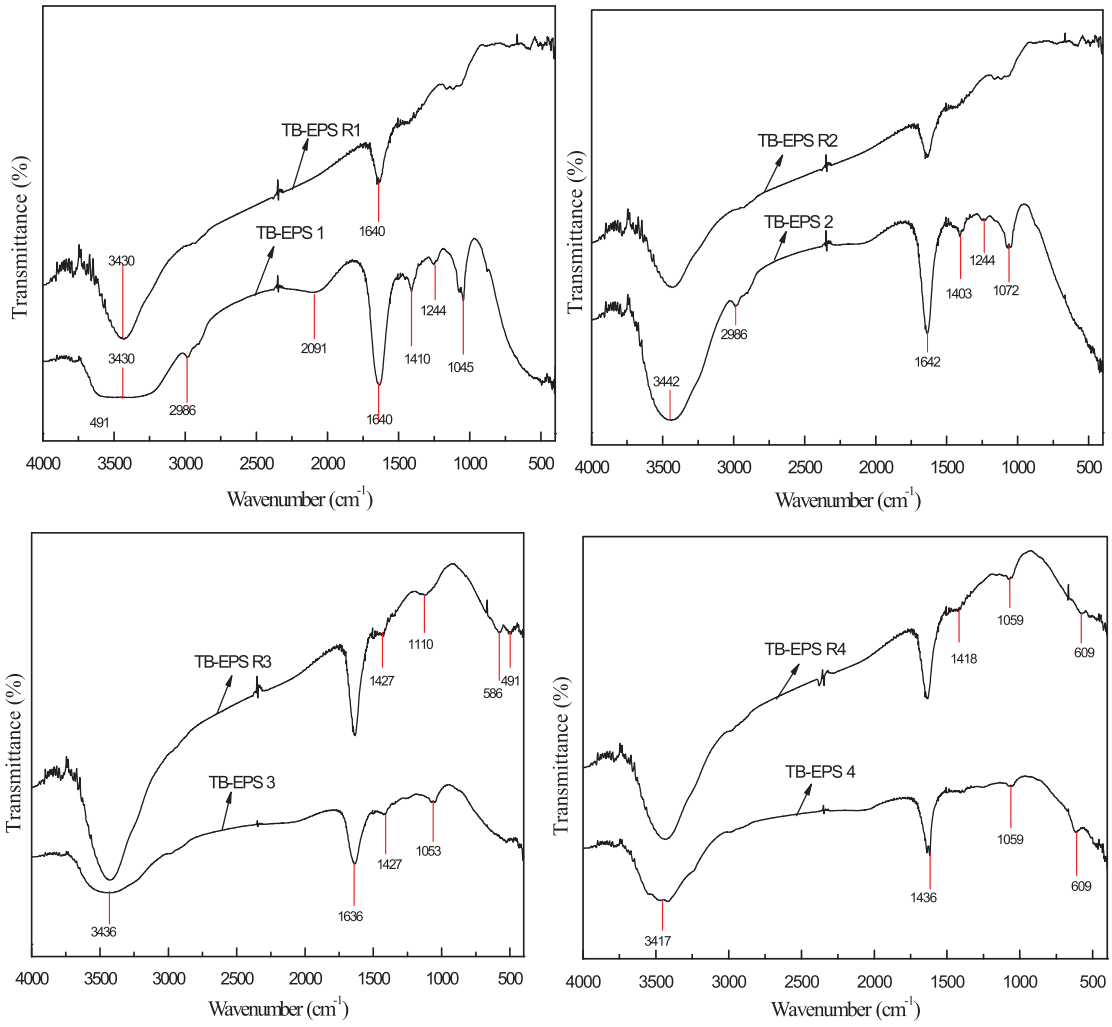


Figure 5. FTIR spectra of TB-EPS in each compartment.

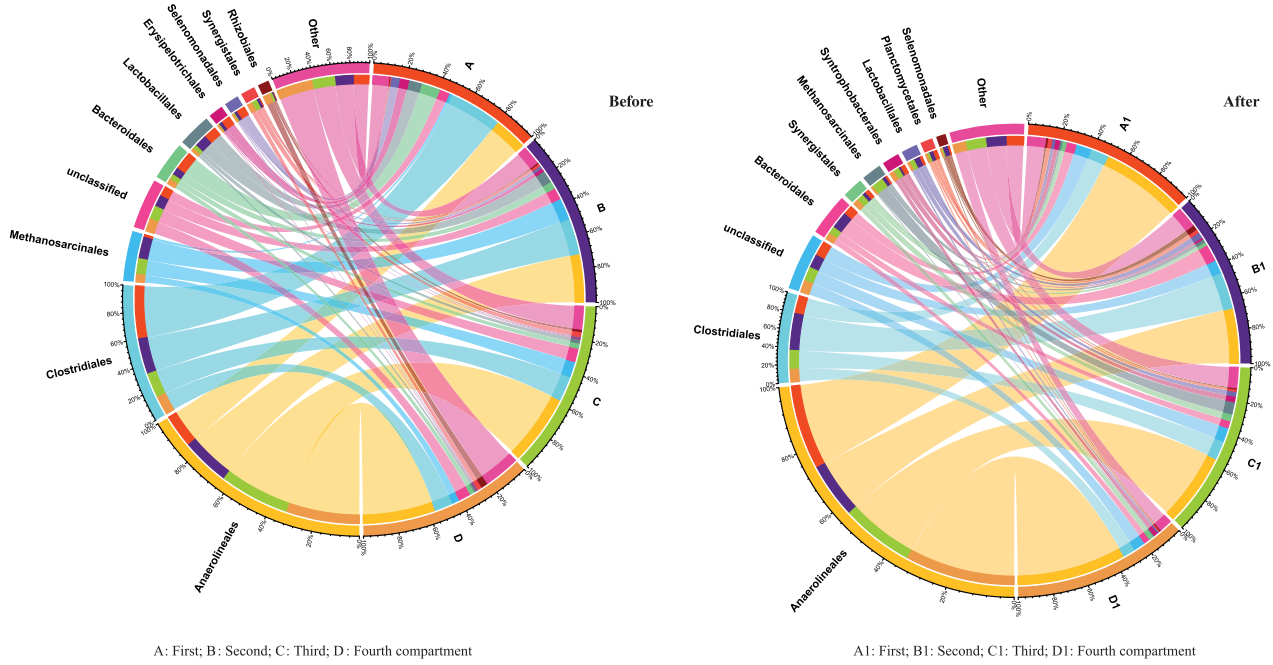
As shown in Fig. S3, the C–H stretching vibration appeared at 2979 cm^{-1}, the N–H bending vibration appeared at 1567 cm^{-1}, and the C–O–C stretching vibration of polysaccharides appeared at 1120–1020 cm^{-1} in the FTIR spectra of the LB-EPS. However, the stretching vibration of C=O at 1647 cm^{-1} and the vibration absorption peak of C=O aromatic compound at 1458 cm^{-1} decreased significantly. These results showed that aromatic compounds decreased and polysaccharides increased in the first compartment. In the third compartment, the peak of N-H stretching vibration (3500–3300 cm^{-1}) was slightly wide and the bending vibration of N-H appeared at 1581 cm^{-1}. The C–N stretching vibration of this amide group at 1655 cm^{-1} and the C–N stretching vibration at 1386 cm^{-1} were predominantly reduced. It is indicated that the modified methods may lead to protein dissolution in microbial cells, which is consistent with the increase of protein-like substances measured by the EEM spectra of LB-EPS. In the fourth compartment, the peak of the N-H vibration (3500–3300 cm^{-1}) was significantly enhanced by the modification. And the peaks of C–H stretching vibration (2970 cm^{-1}), N–H bending vibration (1584 cm^{-1}), and C–O–C stretching vibration (1120–1020 cm^{-1}) appeared in the FTIR spectra. It was demonstrated that aeration had influence on the strict anaerobic bacterial communities in the fourth compartment of the reactor.

For the TB-EPS of the FTIR spectra (Figure 5), the N–H stretching vibration peak (3500–3300 cm^{-1}) changed from wide peak to sharp peak, and the C–H stretching vibration appeared at 2986 cm^{-1} in the first compartment by the modification. However, the C=O stretching vibration peak at 1410 cm^{-1}, the C–N stretching vibration peak at 1640 cm^{-1} and the polysaccharide C–O–C stretching vibration peak at 1120–1020 cm^{-1} significantly reduced or disappeared. The results showed that the modification had a significant effect on the TB-EPS in the sludge of the first compartment, and some polysaccharide substances reduced in the TB-EPS. The

changes of FTIR spectra of the first and second compartment were similar. The N–H stretching vibration peak (3500–3300 cm^{-1}) in the third compartment changed from a wide peak to a sharp peak. At the same time, the stretching vibration area of C=O (1636 cm^{-1}) changed from a short peak to a long peak. The change of the FTIR spectra of TB-EPS in the fourth compartment was not clear. The absorption peak of aromatic compounds at 1400–1500 cm^{-1} increased slightly. It was demonstrated that the modification had a little effect on the TB-ESP of the fourth compartment of the reactor. High aromatic compounds of EPS could promote the flocculation ability and structural stability of sludge due to their unique charge property and three-dimensional scaffold structure, and the variation of aromatic compounds demonstrated that proteins were main composition in the TB-EPS, which could further explain the results of EEM [27,29].

3.3. *Microbial community structure*

After quality filtering, the total number of clean sequence reads was 40,116, 406,616, 49,592, and 53,742 for the first, second, third, and fourth compartment of the reactor before modification, respectively, and the Shannon indices for each of these were 4.79, 4.77, 4.70, and 4.93, respectively. After modification reflux and micro-aeration, the clean number of sequences was 45,339, 45,136, 53,881, and 48,373, respectively. Moreover, the Shannon index of the second and third compartments increased to 5.08 and 4.78, respectively, while the Shannon index of the fourth compartment reduced to 4.05. The diversity of bacterial community compositions reduced in comparison with original sewage sludge. In this study, high throughput sequencing was used to analyse the microbe community dynamic at the level of phylum, order, and genus in the modified ABR. The change in the dominant bacterial communities in the reactor could be identified, which was helpful to understand the operation state of the reactor. Fig. S4 and Fig. S5 are network diagrams at the phylum level before and after modification, respectively. The network diagram reflects species abundance among different microbial communities. The A, B, C, and D samples were respectively expressed as the first, second, third, and fourth compartments before modification. Overall, before modification, *Chloroflexi* and *Firmicutes* were dominant. However, after modification (A1, B1, C1, and D1 samples), the relative abundance of *Firmicutes* was significant decreased and *Chloroflexi* increased. *Chloroflexi* is a facultative anaerobic. The increase of reflux and micro-aeration was beneficial to the growth of *Chloroflexi*. However, *Firmicutes* had three classes of anaerobic, facultative, or obligate aerobic microbes. Therefore, growth of *Firmicutesis* was affected by adding reflux and micro-aeration, resulting in the decrease of relative abundance of *Firmicutes* [30]. *Proteobacteria* and *Firmicutes* in simultaneous denitrification and methanogenesis granular sludge were found in the previous reports [31]. However, after modification,



A: First; B: Second; C: Third; D: Fourth compartment

A1: First; B1: Second; C1: Third; D1: Fourth compartment

Figure 6. Circos of bacterial community at order level.

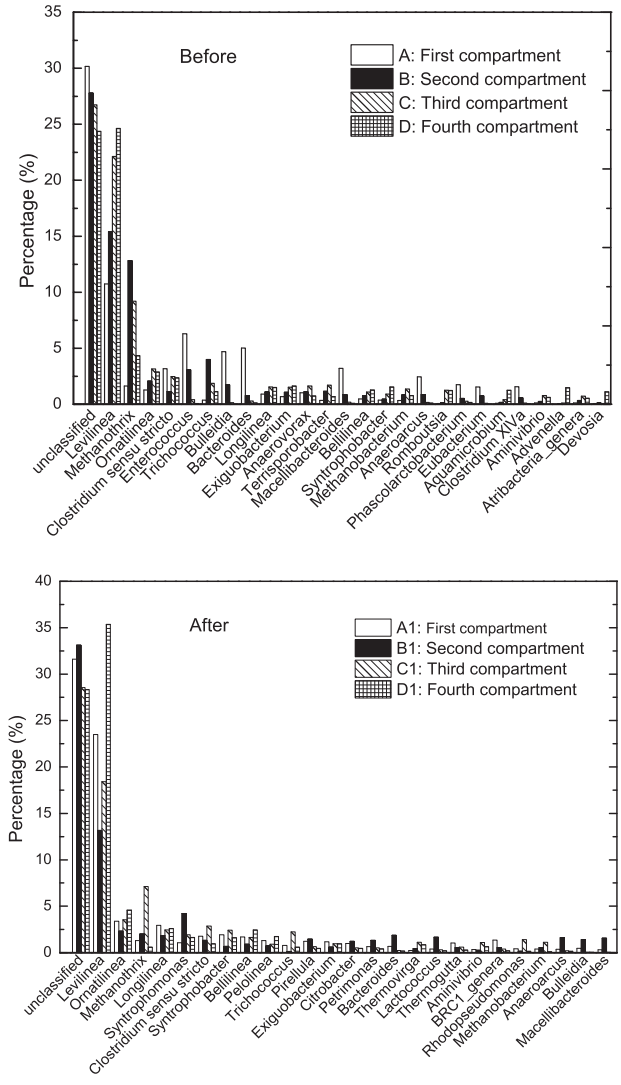


Figure 7. Bacterial community structures at the genus level.

the removal rate of ammonia nitrogen increased, indicating that anammox rather than denitrification was carried out in the reactor. *Synergistetes* contribute to the degradation of long-chain fatty acids, butyrate, and propionate in reactors [9]. The members of these genera are reported to be strictly anaerobic, having ability to utilize a variety of sugars and produce acetate and butyrate [30].

The circos diagram between samples and species before and after the improvement was illustrated on the order level in Figure 6. A circos diagram is a visual circle diagram that can be used to describe the corresponding relationship between samples and microbial communities, and to reflect the proportional distribution of dominant bacteria in each sample. After the modification, the relative abundance of *Anaerolineales* increased in each compartment. The first and fourth compartments had the most significant changes, both increased by 20%. The modification of reflux and micro-aeration was favourable to the growth of *Anaerolineales*. *Anaerolineales* were found in anammox reactors and they played an important role in biodegradation of carbohydrates and other cellular materials (such as amino acids) in anaerobic reactor [30,32]. The relative abundance of *Clostridiales* increased by 20% in the first compartment, but *Clostridiales* decreased by 10% in the third and fourth reactor compartments. The relative abundance of *Erysipelotrichales* and *Rhizobiales* were significantly decreased, while the relative abundance of *Planctomycetales* and *Syntrophobacterales* increased. *Planctomycetales* played a critical role in the anaerobic bioreactor and could be more potential for anaerobic ammonia nitrogen in anammox reactors [33].

As shown in Figure 7, after the modification, *Levilinea* had a significant change in each compartment, with the first compartment increasing by 13%, the second compartment by 2%, the third compartment by 4%, and the fourth compartment by 10.5%. *Syntrophomonas* increased by 5.2% in the first compartment. *Levilinea* and *Syntrophomonas* commonly exist in normal anaerobic digestion systems [34]. *Levilinea* is suggested to play a role in hydrolytic fermentation, and *Levilinea* is resistant to industrial waste water poisoning [35]. *Syntrophomonas* perform syntrophic functions [35]. While the abundance of *Clostridium* decreased from 2.4% to 0.96% in the fourth reactor compartment. *Clostridium* plays a role in syntrophic acetate oxidation and fermentation of various carbohydrates [30]. In addition, the relative abundance of *Methanothrix* reduced by 10.8% in the second compartment, 2.1% in the third compartment, and 3.7% in the fourth compartment. The reason is likely due to the fact that *Methanothrix* belongs to an extremely strict anaerobic bacterial group. *Methanothrix* are known to use acetate as the substrate for methanogenesis. When the ABR had an additional with micro aeration device, the concentration of dissolved oxygen increased and *Methanothrix* was negatively affected in the reactor.

4. Conclusions

The results showed that the modification to add reflux and micro-aeration devices in the ABR had a significant impact on the removal efficiency of NH_3-N and the microbial community. The protein-like substances of sludge in each compartment increased, while the humic acid and aromatic compounds decreased in each compartment of the modified ABR. After the improvement, the relative abundance of *Planctomycetales* and *Syntrophobacterales* increased, while *Methanothrix* was negatively affected in the reactor. This research will be helpful for the operation and useful for improving the performance of ABR.

Disclosure statement

No potential conflict of interest was reported by the authors.

Funding

The study was financially supported by the National Natural Science Foundation of China [grant number 51768009]. We also thank the Doctor Started of Guangxi Normal University [grant number 2016BQ003] and the Natural Science Foundation of Guangxi Province [grant number 2017GXNSFAA198277] for the partial support of this study.

References

[1] Lv LY, Li WG, Wu CD, et al. Microbial community composition and function in a pilot-scale anaerobic-anoxic-aerobic combined process for the treatment of traditional Chinese medicine wastewater. Bioresour Technol. 2017;240:84–93.

[2] He HJ, Zhang ZM, Yang CP, et al. Treatment of organic wastewater Containing high concentration of sulfate by crystallization-Fenton-SBR. J Enviro Eng. 2018;144:04018041.

[3] Peng ZX, He HJ, Yang CP, et al. Biological treatment of wastewater with high concentrations of zinc and sulfate ions from zinc pyrithione synthesis. Trans Nonferrous Met Soc China. 2017;27:2481–2491.

[4] Tang GM, Jia ZH, Yu XM, et al. New advanced treatment of biologically treated effluents from traditional Chinese medicine wastewater using the coupling process of O_3/H_2O_2-BAF. J Taiwan Inst Chem E. 2015;56:167–173.

[5] Aqaneghad M, Moussavi G. Electrochemically enhancement of the anaerobic baffled reactor performance as an appropriate technology for treatment of municipal wastewater in developing countries. Sustain Environ Res. 2016;26:203–208.

[6] Rahman WU, Khan MD, Khan MZ, et al. Anaerobic biodegradation of benzene-laden wastewater under mesophilic environment and simultaneous recovery of methane-rich biogas. J Environ Chem Eng. 2018;6:2957–2964.

[7] Liu XL, Ren NQ, Yuan YX. Performance of a periodic anaerobic baffled reactor fed on Chinese traditional medicine industrial wastewater. Bioresour Technol. 2009;100:104–110.

[8] Fujihira T, Seo S, Yamaguchi T, et al. High-rate anaerobic treatment system for solid/lipid-rich wastewater using anaerobic baffled reactor with scum recovery. Bioresour Technol. 2018;263:145–152.

[9] Li JZ, Shi E, Antwi P, et al. Modeling the performance of an anaerobic baffled reactor with the variation of hydraulic retention time. Bioresour Technol. 2016;214:477–486.

[10] Ban QY, Li JZ, Zhang LG, et al. Linking performance with microbial community characteristics in an anaerobic baffled reactor. Appl Biochem Biotechnol. 2013;169:1822–1836.

[11] Sayedin F, Kermanshahi A, He SQ. Anaerobic digestion of thin stillage of corn ethanol plant in a novel anaerobic baffled reactor. Waste Manage. 2018;78:541–552.

[12] Liu YY, Huang LH, Dong GH, et al. Enhanced granulation and methane recovery at low load by downflow sludge circulation in anaerobic treatment of domestic wastewater. Bioresour Technol. 2018;249:851–857.

[13] Liu X, Dong CJ. Simultaneous COD and nitrogen removal in a micro-aerobic granular sludge reactor for domestic wastewater treatment. Syst Eng Proc. 2011;1:99–105.

[14] Zhang DQ, Zhou Y, Bugge TV, et al. Soluble microbial products (SMPs) in a sequencing batch reactor with novel cake filtration system. Chemosphere. 2017;184:1286–1297.

[15] Zhang DQ, Trzcinski AP, Kunacheva C, et al. Characterization of soluble microbial products (SMPs) in a membrane bioreactor (MBR) treating synthetic wastewater containing pharmaceutical compounds. Water Res. 2016;102:594–606.

[16] Zhang WJ, Cao BD, Wang DS, et al. Influence of wastewater sludge treatment using combined peroxyacetic acid oxidation and inorganic coagulants re-flocculation on characteristics of extracellular polymeric substances (EPS). Water Res. 2016;88:728–739.

[17] Wang BB, Liu XT, Chen JM, et al. Composition and functional group characterization of extracellular polymeric substances (EPS) in activated sludge: the impacts of polymerization degree of proteinaceous substrates. Water Res. 2018;129:133–142.

[18] Su CY, Li WG, Lu YX, et al. Effect of heterogeneous Fenton-like pre-treatment on anaerobic granular sludge performance and microbial community for the treatment of traditional Chinese medicine wastewater. J Hazard Mater. 2016;314:51–58.

[19] Yang B, Xu H, Wang JF, et al. Bacterial and archaeal community distribution and stabilization of anaerobic sludge in a strengthen circulation anaerobic (SCA) reactor for municipal wastewater treatment. Bioresour Technol. 2017;244:750–758.

[20] Wang T, Huang ZX, Ruan WQ, et al. Insights into sludge granulation during anaerobic treatment of high-strength leachate via a full-scale IC reactor with external circulation system. J Environ Sci. 2018;64:227–234.

[21] Chen CJ, Sun FQ, Zhang HQ, et al. Evaluation of COD effect on anammox process and microbial communities in the anaerobic baffled reactor (ABR). Bioresour Technol. 2016;216:571–578.

[22] Zhu L, Qi HY, Lv ML, et al. Component analysis of extracellular polymeric substances (EPS) during aerobic sludge granulation using FTIR and 3D-EEM technologies. Bioresour Technol. 2012;124:455–459.

[23] Shi YH, Huang JH, Zeng GM, et al. Exploiting extracellular polymeric substances (EPS) controlling strategies for performance enhancement of biological wastewater treatments: an overview. Chemosphere. 2017;180:396–411.

[24] Chen YJ, He HJ, Liu HY, et al. Effect of salinity on removal performance and activated sludge characteristics in sequencing batch reactors. Bioresour Technol. 2018;249:890–899.

[25] He HJ, Chen YJ, Li X, et al. Influence of salinity on microorganisms in activated sludge processes: A review. Int Biodeter Biodegr. 2017;119:520–527.

[26] Wang ZP, Zhang T. Characterization of soluble microbial products (SMP) under stressful conditions. Water Res. 2010;44:5499–5509.
[27] Ma BR, Li SS, Wang S, et al. Effect of Fe_3O_4 nanoparticles on composition and spectroscopic characteristics of extracellular polymeric substances from activated sludge. Process Biochem. 2018;75:212–220.
[28] Yan Z, He HJ, Yang CP, et al. Biodegradation of 3,5-dimethyl-2,4-dichlorophenol in saline wastewater by newly isolated penicillium sp. yz11-22N2. J Environ Sci. 2017;57:211–220.
[29] Wang XL, Zhang L, Peng YZ, et al. Enhancing the digestion of waste activated sludge through nitrite addition: insight on mechanism through profiles of extracellular polymeric substances (EPS) and microbial communities. J Hazard Mater. 2019;369:164–170.
[30] Gulhane M, Pandit P, Khardenavis A, et al. Study of microbial community plasticity for anaerobic digestion of vegetable waste in anaerobic baffled reactor. Renew Energ. 2017;101:59–66.
[31] Luo G, Li J, Li Y, et al. Performance, kinetics behaviors and microbial community of internal circulation anaerobic reactor treating wastewater with high organic loading rate: role of external hydraulic circulation. Bioresour Technol. 2016;222:470–477.
[32] Song W, Wang XJ, Gu J, et al. Effects of different swine manure to wheat straw ratios on antibiotic resistance genes and the microbial community structure during anaerobic digestion. Bioresour Technol. 2017;231:1–8.
[33] Liu ST, Yang FL, Meng FF, et al. Enhanced anammox consortium activity for nitrogen removal: Impacts of static magnetic field. J. Biotechnol. 2008;138:96–102.
[34] Kong Z, Li L, Li YY. Characterization and variation of microbial community structure during the anaerobic treatment of N, N-dimethylformamide-containing wastewater by UASB with artificially mixed consortium. Bioresour Technol. 2018;268:434–444.
[35] Antwi P, Li JZ, Boadi PO, et al. Efficiency of an upflow anaerobic sludge blanket reactor treating potato starch processing wastewater and related process kinetics, functional microbial community and sludge morphology. Bioresour Technol. 2017;239:105–116.